

Philosophy Engineering Is Not a Paradigm

A test against Kuhn’s criteria, and a present-tense answer to
Feyerabend’s objection to Lakatos

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Abstract

Philosophy Engineering has been described by its author as a new paradigm for science in Kuhn’s sense. This paper tests that description against Kuhn’s own criteria and rejects it. The programme supplies symbolic generalizations and a changed account of what counts as a solved problem, and it has a small stock of exemplars. It has no community, which for Kuhn is not a rhetorical shortfall but the unit of analysis. It also possesses a specified procedure for translating a legacy corpus into its own vocabulary, and the existence of that procedure is evidence against incommensurability and therefore against revolution. The anomaly the programme names, representation dependence, cannot produce a Kuhnian crisis, because the reigning instruments score a representation-dependent system as correct. Kuhn requires a paradigm to fail by its own standards, and this one does not. What survives the negative result is smaller and better supported. Lakatos distinguished a programme’s hard core from its protective belt and was criticized by Feyerabend for an appraisal available only in hindsight. A claim ledger that fixes priority by repository ancestry and requires dependencies to be declared before the test turns that partition into a property of the record rather than a reconstruction, so the question of what a refutation strikes acquires a present-tense answer. That is not a paradigm. It is a change to the medium in which paradigms are recorded, and the paper states four predictions by which it can be shown wrong.

Keywords: Kuhn, Lakatos, Feyerabend, research programmes, scientific infrastructure, pre-registration, philosophy engineering, claim ledger

1 Introduction

Philosophy Engineering is the practice of taking a philosophical commitment and building something that mechanically refuses to violate it. Its author has described it as a new paradigm for science in the sense of Kuhn (Kuhn, 1970). This paper tests that description. The result is negative.

The test is worth running because the claim is cheap. Announcing a paradigm shift costs nothing, cannot be refuted by any short argument, and is made routinely by programmes that turn out to be ordinary. A discipline whose stated rule is that a programme may not assert what its ledger cannot show is poorly placed to make the cheapest claim in science and leave

it unexamined. The appropriate response is to state the criteria in advance, apply them, and report what happens, including to the parts of the claim that fail.

Three results follow. First, the programme fails Kuhn’s community criterion by a wide margin, and it fails the incommensurability criterion in a way that counts in its favour rather than against it, because the thing that fails the criterion is a working translation procedure. Second, the anomaly on which the strongest case rests cannot do the work Kuhn’s account requires of anomalies, since a system exhibiting it still passes the measurements the reigning approach uses. Kuhn’s crises begin when a paradigm fails by its own standards, and this anomaly is invisible to those standards rather than damaging to them. Third, the programme contains two arms with different verdicts, and the appearance of a paradigm comes largely from running them together: the judgment arm has the shape of a candidate theory change inside one subfield, while the inquiry arm is not a theory at all and was never a candidate.

What remains after the negative result is a smaller claim with better support. Lakatos separated a research programme’s hard core from its protective belt and argued that refutations ordinarily strike the belt (Lakatos, 1970). Feyerabend objected that the resulting appraisal was available only in hindsight, which made it useless to a scientist deciding what to do now (Feyerabend, 1975). The objection has force because nothing in the ordinary scientific record fixes which commitments were core before the test was run, so the partition is reconstructed afterwards by someone with an interest in the answer. A record that fixes priority by repository ancestry, requires dependencies to be declared before the seal, and computes the consequences of a refutation by traversing a graph, turns that partition into something the record already contains. The question of what a refutation strikes then has an answer in the present tense. This is a contribution to the appraisal of research programmes, not a revolution, and the paper states four ways it could be shown wrong.

2 The claimant is badly placed to assess the claim

Kuhn’s own account makes a participant’s declaration of a paradigm shift weak evidence. Revolutions in *Structure* are described as invisible from the inside. The community’s textbooks are rewritten so that the current position appears to be the cumulative result of what came before, and the practitioners who perform the rewriting do not experience themselves as doing so (Kuhn, 1970, ch. 11). If revolutions are hard to see while standing in them, the person best placed to see one is a historian at some distance, and the person worst placed is the programme’s founder.

There is a second difficulty. The word is used loosely. Masterman counted twenty-one distinct senses of “paradigm” in the first edition of *Structure* (Masterman, 1970), and Kuhn accepted enough of the criticism to separate two of them in later work: the disciplinary matrix, meaning everything a community shares, and the exemplar, meaning a concrete problem solution that trains practitioners to recognize new problems as being of the same kind (Kuhn, 1974). A claim to be a paradigm is therefore ambiguous until the sense is fixed. Both senses are tested below.

The procedural safeguard adopted here is the one the programme applies to itself. Under PE-CLS-1.0 no document may cite a claim above its evidence class (Bond, 2026b). The thesis of this paper is supported by one programme, observed by its own author, and has not been tried elsewhere. Its class is **exploratory**, and no other document in the corpus may cite it as more than that. Section 10 states what the record cannot currently show.

3 What Kuhn requires

Kuhn’s criteria are stated here in the form in which they will be applied.

The disciplinary matrix. A paradigm in the broad sense is what the members of a scientific community share. Kuhn decomposes it into symbolic generalizations deployed without argument, models and metaphysical commitments that supply preferred analogies, values that govern what counts as a good solution, and exemplars (Kuhn, 1970, Postscript).

Exemplars. In the narrow sense a paradigm is a concrete problem solution, transmitted through training, that lets a practitioner see an unfamiliar situation as similar to a familiar one (Kuhn, 1974). Kuhn treats this as the primary sense. Exemplars are numerous, they are taught, and they do their work by resemblance rather than by rule.

The community as the unit. Kuhn’s definition is openly circular. A paradigm is what a scientific community shares, and a scientific community consists of those who share a paradigm (Kuhn, 1970, Postscript). The circularity is not an embarrassment in his account; it fixes the unit of analysis. Paradigms are properties of communities of practitioners. They are not properties of positions, theories, or individuals.

Normal science and anomaly. Normal science is puzzle-solving conducted inside the matrix, where the existence of a solution is assumed and the practitioner’s skill is at issue. An anomaly is a result the paradigm cannot assimilate. Anomalies alone do not produce crises. A crisis arises when the paradigm repeatedly fails at puzzles it has itself certified as solvable, and the failure becomes visible to the community using the paradigm’s own instruments (Kuhn, 1970, chs. 6–8).

Incommensurability. Rival paradigms lack a neutral vocabulary in which their claims can be compared without loss. Kuhn’s later statements narrow this to taxonomic incommensurability, a failure of intertranslatability among the kind terms of the two theories. The relevant consequence for this paper is negative and sharp: there is no algorithm that converts the claims of one paradigm into the claims of another with their evidential standing intact. Where such a procedure exists, the two bodies of work are not separated by a revolution.

4 The case for

The case that Philosophy Engineering is a paradigm is stronger than its brevity here suggests, and it should be put at full strength before it is tested.

The programme names an anomaly. Systems deployed in medicine, law, hiring, and content moderation change their verdicts when the description of a case changes in ways that preserve its content. Order of presentation, party names, active against passive voice, choice of synonym, and units of measure all move outputs that ought to be fixed by the situation rather than by its wording. The Foundation document collects the failure modes and treats them as instances of a single defect rather than as unrelated bugs (Bond, 2026a).

The programme supplies symbolic generalizations. The Epistemic Invariance Principle requires a judgment procedure to return equivalent outputs for representations related by a declared transformation group. The Bond Invariance Principle requires that transformations preserving the relational structure of a case leave the evaluation unchanged, and yields an accountability theorem: any two differing judgments differ because the relational structure differs, because the evaluative parameters differ, or both, with no fourth option (Bond, 2026a). Both are stated formally and proved.

The programme changes what counts as a good solution. Under the reigning approach a classifier is good when its aggregate accuracy is high. Under this one a judgment procedure is good when it is invariant under the declared group and still discriminates between cases that differ. Invariance alone is satisfied perfectly by a constant function, so the second condition is not decoration. A change in the criteria of adequacy is exactly what Kuhn lists among the components of a disciplinary matrix.

The programme has exemplars. A failed invariance test produces a witness, a minimal transformation that flips the verdict, and learning to read a witness teaches a practitioner what kind of object a representation-dependence failure is. The derivation of the Klein four-group from Hohfeld’s jural relations (Hohfeld, 1913) shows how a deontic vocabulary becomes a symmetry claim. The withdrawal episode recorded in the corpus shows how a distinction between unsupported and refuted operates on a dependency graph (Bond, 2026b). Brownfield conversion shows what to do with a corpus written before the method existed (Bond, 2026c). These are concrete problem solutions of the kind Kuhn describes, and a practitioner who has worked through them sees new cases differently.

Finally, the field the programme addresses talks like a field in crisis. Its practitioners describe alignment as unsolved, its evaluation methods as saturated, and its failures as poorly understood. Crisis rhetoric is not crisis, but Kuhn treats the community’s own sense of malaise as one of the marks.

5 The test

Kuhn’s criterion	Verdict	Basis
Symbolic generalizations	Partial	EIP and BIP are stated and proved. Kuhn’s generalizations are deployed by a community without argument; these are argued for by one author.
Values and criteria of adequacy	Pass	Invariance with discrimination replaces aggregate accuracy. This is a change in what counts as a solved problem.
Exemplars	Partial	Five or six exist. Kuhn’s paradigms supply hundreds, transmitted by training. There is no training pipeline.
Community	Fail	One author, one department, a small number of students, a chat server. The unit of analysis is absent.
Normal science	Weak	Puzzle-solving inside the matrix occurs, set and solved by the same person.
Crisis in a predecessor	Fail	The anomaly does not register as failure on the reigning instruments. See below.
Incommensurability	Fail	A specified translation procedure exists and has been run. See below.

Community. This is the decisive failure and it does not require elaboration. Kuhn’s paradigms are what communities share. The programme has one author, a handful of student

contributors, and an open chat server. A position held by one person and explained to others is a position. It may be correct. It is not a disciplinary matrix, because there is no discipline whose matrix it is. No amount of formal development substitutes for this, and the criterion is not one a programme can satisfy by wanting to.

The anomaly cannot produce a crisis. This is the more interesting failure, and it follows from the strongest card in the case for.

Kuhn's crises are generated internally. A paradigm certifies certain puzzles as solvable, practitioners attempt them using the paradigm's instruments, and the attempts fail repeatedly in ways the practitioners can see. Representation dependence does not have this structure. A model that returns different verdicts for two descriptions of the same case is scored on a benchmark that contains one of the two descriptions. On that benchmark it is correct. Aggregate accuracy is insensitive to the defect by construction, because accuracy is computed against a fixed encoding of each case and never presents the alternative encoding. The reigning instruments therefore do not register the anomaly as a failure. They do not register it at all.

This is a real defect in current practice, and identifying it is worth doing. But it is the wrong shape for a Kuhnian crisis. Kuhn needs a paradigm that is failing by its own lights, visibly, to the people using it. What is described here is a paradigm succeeding by its own lights at something its critic regards as the wrong target. Kuhn has no category for that, and the reason he has none is that such a complaint cannot build pressure inside the community by the mechanism his account specifies. It has to be argued from outside, in a vocabulary the reigning measurements do not contain. Which is what the programme is in fact doing, and which is not what revolutions look like in *Structure*.

The translation procedure is evidence against revolution. PE-BRW-1.0 specifies how to convert a corpus written under ordinary scholarly norms into a claim ledger (Bond, 2026c). It defines a characterization step that assigns each existing claim the class its actual support warrants, a debt ratio measuring the fraction of claims entered without a prior registration, and a discipline that forbids re-deriving the corpus under the new method, on the ground that priority is the one property no later rigour reconstructs. It has been run. A book of roughly thirty-five thousand lines, six foundation papers, and one further paper converted to 234 claim objects at a debt ratio of 0.962, with five findings and six registrable predictions identified (Bond, 2026f).

Kuhn's incommensurability thesis says no such procedure exists between paradigms. There is no neutral algorithm that carries the claims of one disciplinary matrix into another with their standing intact. PE-BRW-1.0 is an algorithm that does precisely this, it is written down, and its output is inspectable. Two readings are available. Either the corpus and the ledger are not separated by a revolution, or Kuhn is wrong about incommensurability. The first is overwhelmingly more likely.

The right description is that Philosophy Engineering is a conservative extension of ordinary scholarship. It adds constraints on what may be asserted and requires records that were not previously required. It does not change the referents of the terms, and a claim converted into the ledger means what it meant before. The debt ratio near unity is itself the proof: almost every converted claim retained its content and gained only a statement about the evidence behind it. Conservative extensions are valuable. They are not revolutions, and a programme that can translate its predecessor has by that fact shown it is not incommensurable with it.

6 Two arms, two verdicts

The programme has three arms. The judgment arm governs what an autonomous system may assert and contains EIP and BIP. The inquiry arm governs what a research programme may claim and contains the claim ledger. The discovery arm governs what a programme may claim survives and contains the transformation registry (Bond, 2026b,d). Much of the plausibility of the paradigm claim comes from treating these as one thing, and they should be separated.

The judgment arm has the shape of a candidate theory change within one subfield. It supplies generalizations, changes the criteria of adequacy, and has exemplars. Whether it becomes a paradigm inside AI alignment is an empirical question about a community, and the answer is currently no by a wide margin and too early to assess on any other ground. That is the correct thing to say about a programme of this age, and saying more would be the error this paper exists to avoid.

The inquiry arm is not a candidate and never was. It contains no theory of any subject matter. It specifies what a record must contain for certain claims about that record to be checkable. Reform of this kind has a history, and its history is not a history of paradigms. The controlled trial, the journal, peer review, the clinical trial registry, and preregistration each changed what could be claimed and what had to be shown, and none of them is a paradigm in Kuhn's sense. The literature on psychology's credibility revolution has reached the same verdict about that episode: it is substantial reform of standards of evidence rather than a change of theory, and the sources that call it a revolution are using the word in a different sense from Kuhn's (Vazire, 2018).

Combining the two arms produces a rhetorical effect that neither supports alone. The judgment arm lends theoretical content, which reform movements usually lack. The inquiry arm lends scope across all of science, which a subfield theory lacks. A reader who takes both at once perceives something that looks like a general scientific revolution. Assessed separately, one is an early theory in a single field and the other is an infrastructure proposal. Neither is a paradigm.

7 What survives: a present-tense answer to Feyerabend

The negative result leaves a positive claim standing, and it is more defensible than the one it replaces.

Lakatos held that a research programme consists of a hard core its adherents protect by decision and a protective belt of auxiliary hypotheses that absorbs refutations (Lakatos, 1970). A programme is progressive when successive versions predict facts not predicted before and some of those predictions are confirmed, and degenerating when its modifications only accommodate anomalies already known. The account rescues scientific rationality from Popper's difficulty that theories are never abandoned on a single failed prediction, and it does so by moving the unit of appraisal from the theory to the programme.

Feyerabend's objection is that the appraisal arrives too late to be of use. A degenerating programme may recover, so no rule tells a scientist when to abandon one, and the verdict becomes clear only in hindsight (Feyerabend, 1975). The Stanford Encyclopedia states the point in the same terms, calling the methodology pragmatically unsatisfactory because its evaluations are exclusively retrospective and it therefore cannot advise anyone in the present (Musgrave and Pigden, 2025). The objection is usually treated as decisive against Lakatos's

use of the account for demarcation.

The objection depends on a fact about the record rather than a fact about science. The core and the belt are not marked anywhere at the time the predictions are made. After a refutation, someone reconstructs which commitments were central and which were auxiliary, and that person is normally a participant with an interest in the answer. The reconstruction is not dishonest, and it need not be. It is simply unconstrained, because no document written before the test fixed the partition. Kuhn’s observation about textbook rewriting is the same phenomenon at community scale: the record is prose, prose does not carry dependency structure, and what a refutation actually struck cannot be recovered from it.

A claim ledger changes that fact about the record. Three of its properties bear directly on the objection (Bond, 2026b). Priority requires that the commit sealing a registration be an ancestor, in the repository history, of the first commit that adds the result, so a prediction cannot be written after its outcome is known. Coherence requires that every dependency a claim declares resolve to another claim, and that no claim be classified above the weakest support it rests on. Blast radius is then computed by traversing the declared graph, so the set of claims a refutation reaches is obtained by traversal rather than by argument.

The consequence for Lakatos is direct. The core and the belt correspond to position in the dependency graph, with heavily depended-upon claims occupying the core and claims nothing rests on occupying the belt. Because the edges are declared before the test and the ordering is fixed by repository ancestry, the partition is a property of the record at the time of the seal. It is not reconstructed and it cannot be adjusted after the outcome is known without leaving a trace. The question Feyerabend said could only be answered in hindsight, namely what this refutation strikes, acquires an answer available the moment the refutation is recorded.

Three limits on this claim must be stated with it.

First, it addresses one input to Lakatos’s appraisal rather than the appraisal itself. Whether a modification is ad hoc still requires judgment about whether a prediction counted as new, and no graph settles that. The discovery arm’s requirement that transformation families be declared in the registry before the seal constrains the judgment, since a family cannot be selected after seeing which tests passed (Bond, 2026d). It does not eliminate it.

Second, localization reaches only as far as the declared conjunction. Duhem and Quine established that no hypothesis is tested alone, and a refutation indicts whatever was assumed alongside it (Duhem, 1954; Quine, 1951). A ledger can propagate along edges that were written down. Dependencies nobody declared are invisible to traversal, and the specification says so rather than claiming otherwise.

Third, the mechanism assumes the repository history is intact. Priority proved by ancestry is stronger than a third party’s timestamp in requiring no trusted service, and weaker in being vulnerable to history rewriting. The specification records this trade rather than resolving it, and anchoring a hash in an external timestamping service closes the gap cheaply.

8 Predictions

A paper arguing that records govern appraisal should say what would show it wrong. Four predictions follow from the account in §7, stated so that a programme other than this one can test them.

P-I. Blast radii are small and right-skewed. If Lakatos is right that refutations ordinarily strike the belt, and dependencies are declared in advance, then the number of claims

reached by a recorded refutation should be small in most cases with a long tail. Routinely large radii would indicate either that the core and belt picture is wrong or that dependencies are under-declared. The two can be distinguished: under-declaration concentrates large radii on a few claims with high in-degree, while a wrong picture spreads them evenly.

P-II. Recorded negative results rise, then settle high. A programme adopting gap-free identifier assignment should record more null and abandoned lines than matched programmes without a ledger, with the excess concentrated in the first conversion period as previously undisclosed work receives dispositions, followed by a steady state above the matched baseline rather than a return to it.

P-III. Wholesale revision loses its discontinuous character. If the gestalt quality Kuhn attributes to revolutions is partly an effect of a record that cannot represent partial change, then programmes with individually versioned claims should show major revisions as many dated propagations rather than as a single switch. If ledger programmes still revise discontinuously, the record was not the cause and Kuhn’s psychological account survives intact. This is the prediction most likely to fail, and it is the one that most directly tests the paper’s thesis.

P-IV. Declared partitions diverge from retrospective attribution. Take a refutation recorded in a ledger. Ask practitioners who know the field but not the ledger to say which commitments were central before the test. Compare with the declaration sealed beforehand. Kuhn’s invisibility thesis predicts systematic divergence, with retrospective attribution making surviving commitments look more central than they were declared to be. Absence of divergence would indicate that prose records preserve the partition adequately and that the machinery is unnecessary.

P-IV is inexpensive and can be run on the existing corpus. P-I can be computed today from any conforming ledger. P-II and P-III require a second programme, and until a programme other than this one adopts the method, none of the four has been tested.

9 Related work

Nothing in §7 is new in its components, and PRIOR-ART.md states the debts at length (Bond, 2026e). The parts assembled here have the following owners.

Claim-level publication was proposed as nanopublications (Groth et al., 2010) and developed as micropublications with argumentation graphs linking claims to evidence and to each other (Clark et al., 2014). Both predate this specification by a decade or more and both have better serializations. Manuscripts as version-controlled repositories with continuous integration are Manubot’s contribution (Himmelstein et al., 2019). Preregistration established the norm that priority requires a record made before measurement, and registered reports made it a condition of publication (Chambers, 2013). Dependency resolution and deprecation semantics are borrowed wholesale from package management, and proof assistant libraries already compute blast radius exactly for the fragment of claims that can be formalized.

Two adjacent proposals deserve separation. Horstmeyer’s Lakat is a peer-to-peer architecture for continuous integration publishing, explicitly inspired by Lakatos, with a review-based consensus mechanism and a finality rule for branched contributions (Horstmeyer, 2023). Its concern is the governance of a distributed publishing system, and it does not address the appraisal question treated in §7. Vazire’s account of psychology’s credibility revolution describes exactly the change in standards of evidence that the inquiry arm mechanizes, and describes it fifteen

years earlier and with a discipline behind it (Vazire, 2018). The reading offered here, that such reform is not a Kuhnian paradigm shift, is not original either. It is the settled view in that literature.

On infrastructure as a subject for philosophy of science rather than a precondition of it, the relevant work is Bowker and Star on classification systems and their invisible consequences (Bowker and Star, 1999), and Leonelli on how data infrastructures shape what biology can claim (Leonelli, 2016). Hacking’s styles of reasoning (Hacking, 1992) and Galison’s account of trading zones and instrument-driven change (Galison, 1997) both describe scientific change that is not theory change, which is the category this paper places the inquiry arm in.

What is left after these concessions is narrow. The claim is that declaring the dependency partition before the test, and proving priority by repository ancestry, converts Lakatos’s core and belt from a retrospective reconstruction into a property of the record, and thereby answers one standing objection to his methodology. I am not aware of prior art for that specific move. It is also the only part of this paper that would survive if the rest were withdrawn.

We do not offer a better account of scientific revolutions. We offer the test, and the test came out negative.

10 Limits, and what the record cannot show

The programme’s rule is that it may not assert what its ledger cannot show. What follows is the application of that rule to this paper.

One programme, and its author is the subject. Every observation supporting §7 comes from a single corpus, maintained by the person making the argument, who chose the method and then reported that it worked. The confirmation risk is severe and cannot be reduced from inside. None of the four predictions in §8 has been tested against a programme with different authorship.

The checks do not currently pass. Running the discovery arm’s validator against the reference programme fails. Ten tests cite three transformation families the registry never declares. The families were declared in the relevant sealed registrations and were not mirrored into the registry. The finding is left unrepaired, because backfilling a registry after the runs is what the declaration property forbids, and the correct declaration date is not the validator’s to assign (Bond, 2026d). A paper arguing that mechanical checks catch what reading misses should report the case where they caught the author.

Completeness is narrower than it sounds. Gap-free identifier assignment makes the set of identified lines of work enumerable. It says nothing about work that was considered and never given an identifier. The property converts an unfalsifiable claim into a finite check over a bounded domain, and the boundary of that domain is set by the same person the property is meant to constrain.

The corpus contains falsified predictions that were never propagated. Three are recorded: a gauge structure refuted by CHSH-style tests at $N = 600$, a predicted hysteresis effect that failed to confirm and reversed under double-blind conditions at $N = 630$, and a predicted schism effect with no confirmations that failed on the best-powered instrument available. Each was reported where it was found. None had a blast radius computed, because the dependency records did not exist when they were found (Bond, 2026f). That is the pathology §7 describes, observed in the author’s own work before the machinery existed, which is weak evidence that the machinery was needed and no evidence that it succeeds.

This paper is an unregistered claim. It was not sealed before its argument was developed, it has no instrument, and it has not been verified in fresh context. Its class is **exploratory**. Under the assertion authority property, no document in this corpus may cite it above that class, and a reader encountering the thesis of §7 asserted more strongly elsewhere has found a defect worth reporting.

11 Conclusion

The claim tested was that Philosophy Engineering is a new paradigm in Kuhn’s sense. It is not. It has no community, which is the unit Kuhn’s account is about. Its anomaly is invisible to the instruments of the approach it criticizes, so it cannot generate the internal failure a crisis requires. It possesses a working procedure for translating its predecessor’s corpus into its own vocabulary, and the existence of that procedure is evidence that the two are not separated by a revolution at all. The correct description is a conservative extension of ordinary scholarship, which adds requirements about evidence without changing what the claims mean.

The claim that survives is smaller. Lakatos’s separation of a programme’s core from its protective belt was criticized by Feyerabend for yielding verdicts only after the fact, and the criticism holds because ordinary scientific records do not fix the partition in advance. A record that proves priority by ancestry and requires dependencies to be declared before the test makes the partition a property of the record, and what a refutation strikes becomes computable when the refutation is entered rather than arguable afterwards. Section 8 states four ways that account can fail, and until a programme with different authorship adopts the method, none of them has been tested.

There is a reason to prefer the smaller claim beyond its being better supported. Paradigms are announced constantly and the announcements are almost always wrong. Changes to the record are rare, and the ones that took hold, the journal, the controlled trial, the registry, changed what could be claimed without anyone describing them as revolutions. The programme’s own rule applies to its description of itself. It may not assert what its ledger cannot show, and the ledger does not show a paradigm.

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